

Cosmological Constraints from a Joint Analysis of the tSZ Power Spectrum and Cluster Number Counts with Massive-Cluster Masking

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(Dated: July 3, 2026)

We present a self-consistent joint analysis of the thermal Sunyaev-Zel'dovich (tSZ) angular power spectrum and cluster number counts (CNC). Clusters detected above a significance threshold q_{cat} are masked from the power spectrum and counted in CNC, so the two probes act on non-overlapping halo populations. We develop the forward model and likelihood, validate the pipeline on synthetic mock catalogues and painted Compton- y maps (with and without astrophysical foregrounds), and apply the same method to FLAMINGO hydrodynamical simulations.

I. INTRODUCTION

Galaxy clusters probe cosmology through their abundance and through the integrated pressure field traced by the thermal Sunyaev-Zel'dovich (tSZ) effect. Two standard observables are cluster number counts (CNC) and the tSZ angular power spectrum. These probes are complementary but share a practical complication: the most massive, best-detected clusters contribute both to the catalogue and to the low-multipole power spectrum. A naive combination would double-count the same halos.

A natural resolution is to mask detected clusters from the map before measuring the power spectrum, while using the masked objects as the CNC sample. The unresolved tSZ background and the resolved catalogue then probe disjoint halo populations, provided the same selection criterion is used in both legs of the analysis.

This paper is a theory study that develops and demonstrates such a joint pipeline. We proceed in two stages:

1. **Synthetic mocks.** Mock catalogues and painted full-sky Compton- y maps generated under the same forward model as the likelihood. We consider a signal-only case and a foreground-contaminated case.
2. **FLAMINGO simulations.** The same analysis applied to hydrodynamical lightcones, testing the parametric forward model against realistic intracluster-medium physics.

II. METHOD

A. Observables and split halo population

Let q_{obs} denote the detection significance of a cluster. We adopt a catalogue threshold q_{cat} and define

$$\mathcal{S}_{\text{CNC}} = \{\text{halos with } q_{\text{obs}} \geq q_{\text{cat}}\}, \quad (1)$$

$$\mathcal{S}_{\text{PS}} = \{\text{halos with } q_{\text{obs}} < q_{\text{cat}}\}. \quad (2)$$

Clusters in $\mathcal{S}_{\text{CNC}}$ are removed from the Compton- y map (masked) and binned in redshift and significance for CNC. Clusters in $\mathcal{S}_{\text{PS}}$ source the masked tSZ power spectrum. When the map mask and catalogue selection are matched, the two probes are self-consistent and no explicit cross-covariance term is required in the joint likelihood.

B. Compton- y and the tSZ power spectrum

The Compton- y parameter is the line-of-sight integral of the electron pressure,

$$y(\hat{n}) = \frac{\sigma_T}{m_e c^2} \int \frac{P_e dl}{l^2 + d_A^2 \mu^2}, \quad (3)$$

where σ_T is the Thomson cross section, m_e the electron mass, P_e the electron pressure, d_A the angular diameter distance, and μ the angular separation from the cluster centre.

Within the halo model, the one-halo tSZ power spectrum is

$$C_\ell^{\text{tSZ,1h}} = \int dz \frac{dV}{dz d\Omega} \int d \ln M \frac{dn}{d \ln M} |\tilde{y}_\ell(M, z)|^2, \quad (4)$$

where $dn/d \ln M$ is the halo mass function and $\tilde{y}_\ell$ the two-dimensional Fourier transform of the projected Compton- y profile. We report the dimensionless spectrum

$$D_\ell \equiv \frac{\ell(\ell+1)}{2\pi} C_\ell. \quad (5)$$

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For a generalised NFW (GNFW) pressure profile [1],

$$P_e(x) = C(M, z) P_0 \frac{(c_{500}x)^{-\gamma}}{[1 + (c_{500}x)^\alpha]^{(\gamma-\beta)/\alpha}}, \quad x \equiv r/r_s, \quad (6)$$

with $r_s = r_{500c}/c_{500}$ and a mass-redshift-dependent normalisation $C(M, z)$. The central Compton- y amplitude is

$$y_0(M, z) = \frac{2\sigma_T}{m_e c^2} P_{500}(M, z) r_{500c}(M, z) \mathcal{I}_{\text{GNFW}}, \quad (7)$$

where $\mathcal{I}_{\text{GNFW}}$ is the dimensionless integral of the GNFW shape out to the adopted outer radius. We allow a parametric scaling relation

$$y_0^{\text{param}}(M, z) = 10^{A_{\text{SZ}}} \left(\frac{\tilde{M}_{500c}}{3 \times 10^{14} M_\odot} \right)^{\alpha_{\text{SZ}}} E(z)^2 h^{-1/2}, \quad (8)$$

with biased mass $\tilde{M}_{500c} = M_{500c}/B$ and hydrostatic bias $B \equiv 1/(1-b)$.

C. Cluster SNR and detection probability

Following matched-filter cluster detection, the mean signal-to-noise ratio of a halo of mass M at redshift z is

$$q(M, z) = \frac{y_0(M, z)}{\bar{\sigma}_{\text{noise}}(\theta_{500})}, \quad (9)$$

where $\theta_{500} = r_{500c}/d_A$ and $\bar{\sigma}_{\text{noise}}$ is the survey noise at the cluster angular scale.

At fixed (M, z) the tSZ amplitude has intrinsic scatter. We model a log-normal multiplicative factor,

$$\ln A \sim \mathcal{N}(0, \sigma_{\ln Y}^2), \quad y_0 \rightarrow A y_0(M, z), \quad (10)$$

so the pre-noise SNR is $q_m = A \bar{q}(M, z)$. The observed significance includes Gaussian measurement noise,

$$q_{\text{obs}} = q_m + n, \quad n \sim \mathcal{N}(0, 1). \quad (11)$$

Marginalising over intrinsic scatter and measurement noise, the detection probability is

$$P_{\text{det}}(M, z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} du e^{-u^2/2} \Phi\left(\frac{q_m(u) - q_{\text{cat}}}{\sqrt{2}}\right), \quad (12)$$

where $u = \ln A / \sigma_{\ln Y}$, $q_m(u) = \bar{q}(M, z) e^{\sigma_{\ln Y} u}$, and Φ is the standard normal cumulative distribution function.

D. Conditional-moment masking

The masked (unresolved) power spectrum receives contributions only from halos with $q_{\text{obs}} < q_{\text{cat}}$. A halo with scatter factor A contributes $A^2 |\tilde{y}_{\ell,0}|^2$ to the spectrum, so one must average over the *undetected* subpopulation

using the conditional second moment, not the factorised form $\langle A^2 \rangle (1 - P_{\text{det}})$.

We define the conditional n th-moment weight

$$W_n(M, z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} du e^{-u^2/2} e^{n\sigma_{\ln Y} u} \Phi\left(\frac{q_{\text{cat}} - q_m(u)}{\sqrt{2}}\right). \quad (13)$$

The masked one-halo spectrum is then

$$C_\ell^{\text{tSZ, unres}} = \int dz \frac{dV}{dz d\Omega} \int d \ln M \frac{dn}{d \ln M} |\tilde{y}_{\ell,0}(M, z)|^2 W_2(M, z). \quad (14)$$

In the absence of masking, intrinsic scatter boosts the full-sky spectrum by $\langle A^2 \rangle = e^{2\sigma_{\ln Y}^2}$; for the masked case, the mass- and redshift-dependent factor W_2 replaces this uniform boost.

The trispectrum term involves $|\tilde{y}_\ell|^2 |\tilde{y}_{\ell'}|^2 \propto A^4$ and therefore requires W_4 :

$$T_{\ell\ell'}^{\text{unres}} = \int dz \frac{dV}{dz d\Omega} \int d \ln M \frac{dn}{d \ln M} |\tilde{y}_{\ell,0}|^2 |\tilde{y}_{\ell',0}|^2 W_4(M, z). \quad (15)$$

Bin-averaged bandpowers use

$$\bar{T}_{bb'}^{\text{unres}} = \frac{1}{|\mathcal{B}_b| |\mathcal{B}_{b'}|} \sum_{\ell \in \mathcal{B}_b} \sum_{\ell' \in \mathcal{B}_{b'}} T_{\ell\ell'}^{\text{unres}}, \quad (16)$$

where $\mathcal{B}_b$ is the set of integer multipoles in bin b . The masked bandpower covariance is

$$M_{bb'}^{\text{unres}} = \frac{2 (C_{\ell_b}^{\text{unres}} + N_{\ell_b})^2}{(2\ell_b + 1) f_{\text{sky}} \Delta_{\ell_b}} \bar{T}_{bb'}^{\text{unres}} + \frac{4\pi}{f_{\text{sky}}} \delta_{bb'}. \quad (17)$$

E. Cluster number counts

The resolved clusters in $\mathcal{S}_{\text{CNC}}$ are binned in redshift and observed significance. The expected count in bin (i, j) is

$$\lambda_{ij} = 4\pi f_{\text{sky}} \int_{z_i}^{z_{i+1}} dz \frac{dV}{dz d\Omega} \int d \ln M \frac{dn}{d \ln M} \mathcal{P}_{\text{bin}}(q_j, q_{j+1} | M, z), \quad (18)$$

with

$$\mathcal{P}_{\text{bin}}(q_j, q_{j+1} | M, z) = P_{\text{det}}(q_j; M, z) - P_{\text{det}}(q_{j+1}; M, z). \quad (19)$$

The CNC log-likelihood is a product of independent Poisson terms,

$$\ln \mathcal{L}_{\text{CNC}} = \sum_{i,j} [N_{ij}^{\text{obs}} \ln \lambda_{ij} - \lambda_{ij}]. \quad (20)$$

F. Power spectrum likelihood and foreground model

When foregrounds are included (synthetic validation with FG), the model for the observed masked spectrum

is

$$C_\ell^{yy} \simeq C_\ell^{\text{tSZ,unres}} + A_{\text{CIB}} C_\ell^{\text{CIB}} + A_{\text{IR}} C_\ell^{\text{IR}} + A_{\text{RS}} C_\ell^{\text{RS}} + A_{\text{CN}} C_\ell^{\text{CN}}. \quad (21)$$

The power spectrum log-likelihood is Gaussian,

$$-2 \ln \mathcal{L}_{\text{PS}} = (\hat{\mathbf{D}} - \mathbf{D})^\top \mathbf{M}^{-1} (\hat{\mathbf{D}} - \mathbf{D}) + \ln |\mathbf{M}| + \text{const.}, \quad (22)$$

where $\hat{\mathbf{D}}$ and $\mathbf{D}$ are vectors of observed and theoretical bandpowers D_ℓ , and $\mathbf{M}$ is the masked covariance (17) evaluated at the model parameters (with any Hartlap correction applied as needed).

G. Joint likelihood

The joint log-likelihood is

$$\ln \mathcal{L}_{\text{joint}} = \ln \mathcal{L}_{\text{PS}} + \ln \mathcal{L}_{\text{CNC}}. \quad (23)$$

Cosmological parameters, tSZ scaling parameters (A_{SZ} , α_{SZ} , $\sigma_{\ln Y}$, B), and (when relevant) foreground amplitudes are shared between the two probes. Inference is performed with Markov-chain Monte Carlo sampling (e.g. Cobaya).

III. SYNTHETIC DATA

We generate synthetic Planck-like data with a halo-painting pipeline consistent with the likelihood model:

1. Poisson-sample halos from the halo mass function at a fiducial cosmology.
2. Assign intrinsic log-normal scatter factors [Eq. (10)].
3. Paint GNFW profiles on a HEALPix Compton- y map at the scattered amplitudes.
4. Draw observed significances [Eq. (11)].
5. Populate the CNC catalogue above q_{cat} and mask those clusters from the map.
6. Extract full-sky and masked bandpowers $\hat{D}_\ell$ on a fixed binning.
7. Optionally add fixed foreground templates [Eq. (21)] for the FG case.

IV. RESULTS: SYNTHETIC MOCKS WITHOUT FOREGROUNDS

In the signal-only case we compare:

- CNC-only inference,
- masked tSZ PS-only inference,

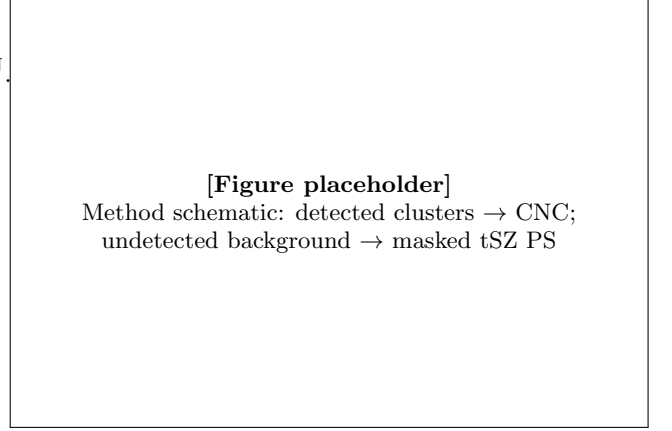


FIG. 1. Cartoon of the joint analysis. Clusters above q_{cat} are counted in CNC and masked from the Compton- y map before the power spectrum is measured.

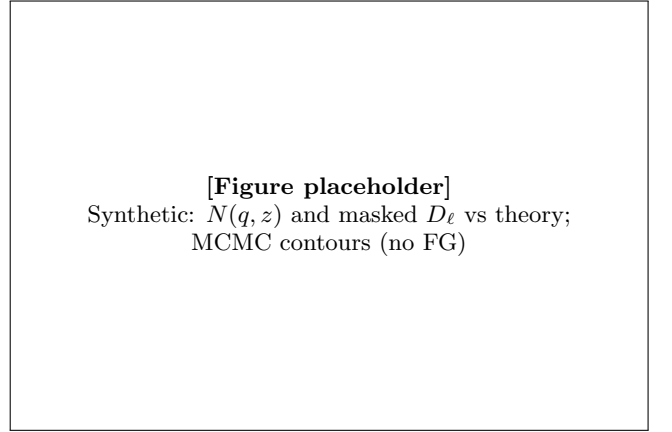


FIG. 2. Synthetic validation without foregrounds. [Caption to be completed.]

- full-sky tSZ PS-only inference (optional control),
- joint CNC + masked tSZ PS,
- joint CNC + full-sky tSZ PS (optional control).

We show agreement between painted maps, mock catalogues, and the forward model for $N(q, z)$ and masked D_ℓ , and compare posterior constraints from the single-probe and joint chains.

V. RESULTS: SYNTHETIC MOCKS WITH FOREGROUNDS

We repeat the analysis with foreground templates added to the painted maps. Without free foreground amplitudes, the PS leg is contaminated and the joint fit is unreliable. With foreground nuisance parameters and informative priors in Eq. (21), the joint analysis can marginalise over FG while CNC anchors the cluster sector.

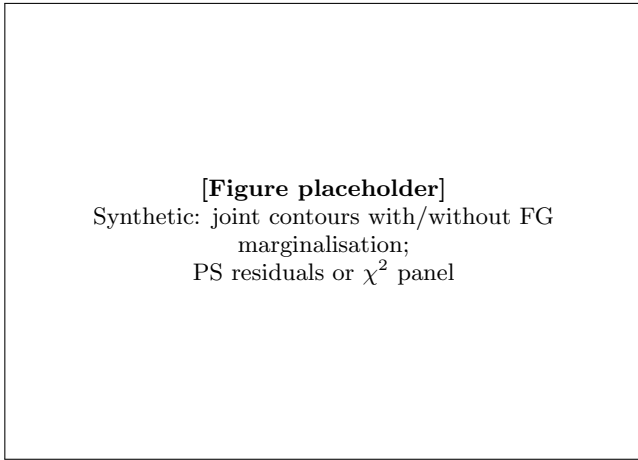


FIG. 3. Synthetic validation with foreground contamination. [Caption to be completed.]

VI. FLAMINGO SIMULATIONS

We apply the same pipeline to FLAMINGO hydrodynamical lightcones. Compton- y maps and SOAP cluster catalogues are built from the L2p8_m9 run (with optional L1_m9 feedback variants for systematics). Cluster significances are assigned with the same GNFW-based forward model and survey noise curve as in the synthetic stage. Detected clusters are masked with apodised discs before bandpowers are estimated (e.g. with NaMaster).

VII. RESULTS: FLAMINGO

We present:

1. Comparison of FLAMINGO $N(q, z)$ and masked D_ℓ to the hmfast / cosmocnc forward model.
2. The same set of single-probe and joint inference runs as in Secs. IV–V, but on FLAMINGO data (signal-only; no astrophysical foregrounds in the simulation maps).
3. Optional: response of CNC and masked PS to L1_m9 feedback prescriptions (jet, f_{gas} , stellar mass threshold variants).

VIII. DISCUSSION

Summarise:

- Validation on synthetic mocks: method works without FG; FG must be constrained when present.
- Application to FLAMINGO: where the parametric model agrees or disagrees with hydrodynamical reality.

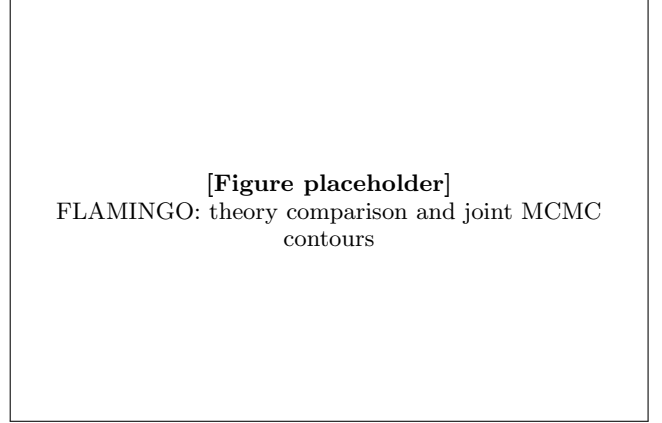


FIG. 4. FLAMINGO application. [Caption to be completed.]

- Role of masking for self-consistency and for making the PS likelihood approximately Gaussian.
- Outlook: real survey data (Planck, Simons Observatory, CMB-S4).

IX. CONCLUSIONS

We have presented a self-consistent framework for joint cluster number counts and masked tSZ power spectrum analysis, validated it on synthetic painted maps, and applied it to FLAMINGO simulations.

ACKNOWLEDGMENTS

We thank [TBD].

[1] M. Arnaud et al., A&A **517**, A92 (2010).